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## EXECUTIVE SUMMARY

Lead Robotics Engineer with an M.S. in Robotics from Georgia Tech and publications in top robotics conferences: ICRA and IROS. Proven experience in leading teams to achieve top-ranked operational performance in large-scale robotic automation systems. Skilled in the full automation stack, from low-level hardware debugging and PLC integration/debugging to high-level AI model research/development and large-scale data analysis.

## EDUCATION

**Georgia Institute of Technology** - M.S. Robotics (AI, CV, Controls), 3.83/4.00 GPA 2021 - 2023

**Tennessee Technological University** - B.S. Mechanical Engineering (Mechatronics), 3.95/4.00 GPA 2015 - 2019

## EXPERIENCE

**Lead Robotics Engineer - Symbotic Automation** 2023 - Present

- Served as the technical lead for a 15+ person team and hundreds of robotic and electrical systems
- Directly contributed to the site becoming #1 site out of 23 in efficiency (cases per man hour)
- Spearheaded the scaling of site throughput by 100x (1k to 100k cases/day) within months, achieving final acceptance (milestone rates and availability) faster than any other site
- Used PLC, software, and electrical tools to repair and debug robotic and electrical systems
- Developed custom Python and SQL automation scripts to analyze large-scale operational data, enabling data-driven debugging and performance optimizations that reduced system downtime and MTTR

**Graduate Research Assistant - Robotics Lab** 2022 - 2023

- Led research projects from conception to publication, resulting in papers at ICRA 2024 and IROS 2023 on multi-task mobile manipulation and visual contact pressure estimation using novel transformer-based models achieving SOTA results
- Learning Robotic Tasks from Video Demonstration - Programmed a robot that, in a simulated robot environment, learns tasks using only video data with a transformer using PyTorch

**Project Lead Robotics Engineer - E.G.O. Products** 2022

- Programmed AGV to store and deliver 500 spools to 4 lines with robust error handling
- Trained 30 workers and 4 engineers to interact with my custom user interface and the robot

**Project Lead Engineer - Johnson Controls** 2020 - 2021

- Worked on a cost-saving material change for 3 different sprinklers with an annual volume of 2 million units
- Designed a sprinkler with a new custom wrench-able cap design and wall bracket

**Mechanical Engineer - Protomet Manufacturing** 2018

- Designed and manufactured a universal speaker mount that has been sold to companies and designed other products

**Engineer - Oak Ridge National Laboratory** 2016

- Worked with fire modeling software (FDS) to discover the optimal building safety design

## PUBLICATIONS

*ForceSight: Multi-Task Text-Guided Mobile Manipulation with Visual-Force Goals* – Links: [Paper](#), [Website](#) ICRA 2024

- Proposes visual location and force goals for mobile manipulation, enabling a variety of robotic tasks

*Visual Contact Pressure Estimation for Grippers in the Wild* – Links: [Paper](#), [Video](#) IROS 2023

- With an image as input, our model achieves SOTA contact pressure and force/torques estimations for robot grippers

## SKILLS

**Languages:** Python, C++, MATLAB, SQL || **Robotics & Software:** ROS2, Git, Linux, Solidworks, CAD, LabVIEW

**AI / ML:** PyTorch, TensorFlow, Computer Vision, Machine Learning, Reinforcement Learning (RL)

**Hardware / Hands-On:** ABB Robots, Automation/Electrical Systems Debugging, PLC (Allen Bradley), Arduino